

MTGeophysics.jl: A software repository for magnetotelluric research and applications

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Summary

The magnetotelluric (MT) method uses natural electromagnetic field variations to image subsurface electrical resistivity at depths ranging from tens of metres to hundreds of kilometres. It is widely used in geophysical exploration for natural resources, including minerals, geothermal energy, and groundwater, as well as in the characterisation of sites for CO₂ storage and in studies of crustal and lithospheric structure. The MT inverse problem is, however, inherently non-unique: many distinct resistivity models reproduce the observations equally well. Conventional practice nonetheless reports a single regularised model, with no quantitative measure of its uncertainty.

MTGeophysics.jl is a Julia package developed to address this limitation. Its principal aim is the quantification of uncertainty in MT model building, advancing interpretation from a single deterministic model, through an ensemble of plausible models, towards the full posterior distribution ([Figure 1](#)). In addition, the package provides modern capabilities for forward modelling, inversion, and visualization in support of state-of-the-art MT research and applications.

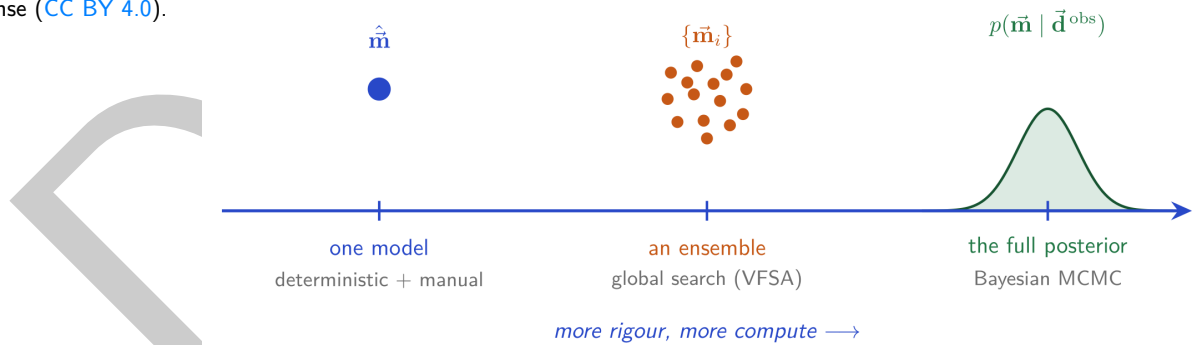


Figure 1: The vision behind MTGeophysics.jl. Deterministic inversion yields a single regularised model $\hat{\mathbf{m}}$; an ensemble of plausible models $\{\mathbf{m}_i\}$ and, ultimately, the full posterior distribution $p(\mathbf{m} | \mathbf{d}^{\text{obs}})$ characterise model uncertainty with increasing rigour and computational cost. The package aims to make ensemble-based uncertainty quantification routine and to provide a practical path towards the full posterior.

Statement of need

Sampling many models consistent with the data, rather than a single regularised model, has been considered impractical in 3D, because a regional mesh carries millions of free parameters and every candidate model requires a full electromagnetic forward solve.

24 MTGeophysics.jl closes this gap in practice. It provides a stochastic inversion workflow based
25 on Very Fast Simulated Annealing (VFSA) (Sen & Stoffa, 2013) that produces an *ensemble* of
26 plausible 3D models, summarised by its mean, median, and standard deviation. The workflow
27 reuses the community-standard ModEM forward solver (Egbert & Kelbert, 2012; Kelbert et al.,
28 2014) by reading and writing its native file formats, so it slots into established MT practice
29 without asking users to change their models, data, or solver.

30 The package targets MT researchers and students who want an open-source toolkit that
31 builds on Julia's (Bezanson et al., 2017) strengths in numerical computing, composability, and
32 interactive graphics. It is designed as a research repository and as a core component of a
33 broader JuliaGeophysics ecosystem: forward solvers, data structures, and inversion routines are
34 meant to be reused and recombined, so that new ideas can be prototyped without rebuilding
35 core MT tooling.

36 State of the field

37 Several open-source tools address parts of the MT workflow. ModEM (Egbert & Kelbert,
38 2012; Kelbert et al., 2014) is the community standard for deterministic 3D MT inversion
39 but provides no built-in visualization or uncertainty quantification. MARE2DEM (Key, 2016)
40 provides adaptive finite-element 2D modelling and inversion of MT and controlled-source
41 electromagnetic data, but does not address 3D problems. MTPy (Kirkby et al., 2019) provides
42 comprehensive Python utilities for MT data handling and visualization but no forward solvers
43 or stochastic inversion. pyGIMLi (Rücker et al., 2017) and SimPEG (Cockett et al., 2015)
44 are general Python inversion frameworks with MT modules, but their MT functionality is
45 embedded in much larger codebases. We are not aware of one that integrates stochastic
46 inversion with ensemble uncertainty quantification in both 2D and 3D. Implementing such a
47 workflow within those frameworks would mean adopting their own meshes and forward solvers,
48 whereas 3D MT models and data in practice are largely held in ModEM formats and inverted
49 with ModEM. A standalone package operating directly on those files, with its own lightweight
50 1D and 2D solvers for rapid experimentation, was therefore the more practical route.

51 Software design

52 The central design choice in MTGeophysics.jl is reduced model parameterisation, the strategy
53 that makes stochastic 3D inversion tractable. Stochastic inversion becomes intractable if every
54 mesh cell is a free parameter: the search space is too large and each proposal requires an
55 expensive forward solve. Rather than perturbing the full grid, the VFSA workflow perturbs a
56 small set of radial-basis-function (RBF) control points and maps them back to the mesh with a
57 compactly supported Gaussian RBF (truncated at 3σ), as shown for one iteration in Figure 2.

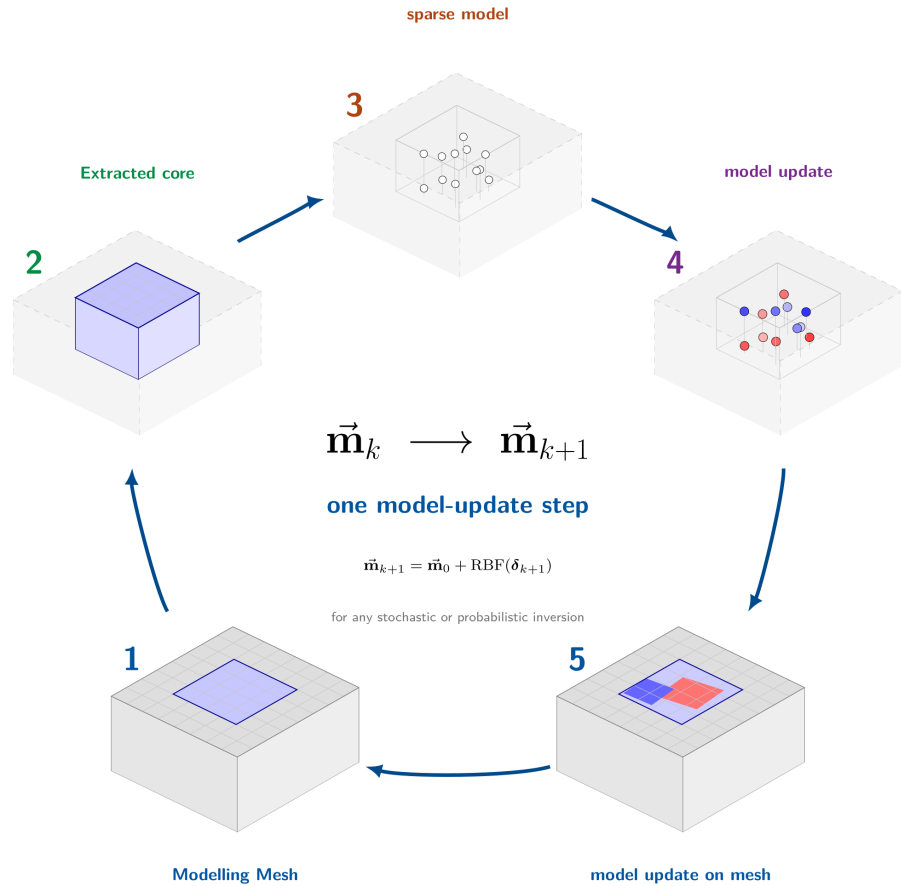


Figure 2: One VFSA iteration as a sparse-model update loop. The padded modelling mesh (1) is reduced to its core (2), sampled at M random control points, the *sparse model* (3), perturbed by the VFSA rule (4), then mapped back to the full padded mesh by compactly supported Gaussian-RBF interpolation (5) for the ModEM forward solve, advancing $\mathbf{m}_k \rightarrow \mathbf{m}_{k+1}$.

58 The same principle extends beyond VFSA: searching a low-dimensional representation is a
 59 prerequisite for practical probabilistic inversion. Fully Bayesian approaches such as Markov
 60 chain Monte Carlo mirror the multi-chain structure already used here, but become affordable in
 61 3D only under a similarly compact model. The roadmap follows the same logic: differentiable
 62 open-source forward solvers, neural surrogates for the expensive 3D solve, and implicit neural
 63 representations that recast the reduced parameterisation as a learned continuous field. That
 64 combination has already been demonstrated for 3D gravity inversion (Mishra, Laaksonen, et
 65 al., 2026), and Julia's scientific-machine-learning ecosystem makes MT the natural next target.

66 Around this core, the package is organised as a single Julia module with clearly separated layers:
 67 ModEM 3D data/model I/O; a 1D module that solves the layered-earth impedance recursion
 68 exactly, with Fréchet derivatives by automatic differentiation; a 2D module that assembles
 69 sparse finite-difference operators for the TE/TM Maxwell equations and solves them with
 70 LinearSolve.jl; and the inversion modules, with Gauss–Newton in 1D and 2D, NLCG in 2D,
 71 and VFSA in 1D, 2D and 3D. In 1D and 2D the package uses its own forward engines and runs
 72 VFSA chains in parallel threads; in 3D it wraps the external ModEM solver, each chain runs as
 73 an independent job, and ensemble statistics are computed across the completed runs. Because
 74 the driver reaches ModEM only through its files and a subprocess call, the forward engine is
 75 swappable. Extending this interface to open-source 3D MT solvers is planned, which would
 76 remove the one component users must currently obtain separately. Interactive visualization

is handled through GLMakie (GPU-accelerated 3D slice viewers with coordinate reprojection via Proj.jl and optional shapefile overlays), while CairoMakie produces publication-quality static plots. Both are regular dependencies; the interactive viewers need OpenGL and are disabled with a warning if GLMakie cannot be loaded, while the static plots need no display. All model, data and control files are plain text for reproducibility and version control, and argument-driven example scripts make every workflow reproducible from the command line. In future, this scriptable, file-based interface is intended to support the development of agentic AI systems, in which AI agents could set up, run, and interpret MT modelling and inversion workflows. Detailed usage (the interactive slice viewers and the polygon-based model editor) is documented in the package documentation (Mishra, 2026).

Correctness is guarded by continuous integration on GitHub Actions. Every push and pull request runs the full test suite on Linux against the two most recent Julia releases (at the time of writing, 1.13 and 1.12), so the package tracks the latest toolchain while the previous release remains supported. Because GLMakie cannot precompile without a display, the workflow provisions a virtual framebuffer (Xvfb) so the visualization layer is exercised headlessly in a clean environment. The suite combines checks of the core 3D data structures and ModEM I/O with analytic checks of the 1D solver, physical consistency and finite-difference Fréchet checks of the 2D solver, the COMMEMI benchmark generators, and end-to-end 1D and 2D inversions (Gauss–Newton, NLCG and VFSA), so solver and inversion behaviour is re-verified on every change. Companion workflows build and deploy the documentation and keep dependency bounds current.

Benchmarks and validation

The package is validated at two levels. The COMMEMI 2D benchmark models (Zhdanov et al., 1997) are generated natively by `helpers/benchmarks_2D.jl`, so the 2D forward and inversion workflows can be exercised from the repository without external downloads. At regional scale, the 3D VFSA workflow has been applied to USArray MT data from the Cascadia subduction zone (Patro & Egbert, 2008) and compared with a deterministic ModEM inversion (Mishra, Kamm, et al., 2026): the ensemble mean recovers the major conductive structures, while the ensemble standard deviation indicates where the data poorly constrain the model. The comparison is reproduced in the package documentation (Mishra, 2026).

Research impact statement

MTGeophysics.jl supports ongoing magnetotelluric research at the Geological Survey of Finland (GTK) and opens uncertainty-aware MT interpretation to the wider community.

The package and the research built on it have been presented to the geophysics and scientific-computing communities by Pankaj K Mishra and Cedric Patzer, including at the EGU General Assembly 2026, the 27th Electromagnetic Induction Workshop (EMIW 2026), and JuliaCon 2026. MTGeophysics.jl was also used to participate in the MT3DINV-4 workshop (Memorial University of Newfoundland, 2025), a community 3D MT inversion benchmarking exercise in which many groups inverted a common real field dataset acquired over the Raglan mining district in northern Quebec, Canada.

AI usage disclosure

Different versions of Claude (Anthropic) and Codex (OpenAI) were used over the duration of development of this package to implement ideas, reorganise the repository, write documentation and tests, and edit this paper. The GitHub Copilot coding agent contributed a small number of commits, and is also used to check for continuous integration (CI) errors on GitHub;

Dependabot is used to keep dependency versions current. All AI-generated code and text were reviewed, tested, and verified by the author for correctness.

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